

Python based Project : phase1-Problem Understanding and Logic Designing

Problem statement: Develop a Python-based Random Password Generator that creates strong, secure, and customizable passwords based on user-defined requirements. The application should allow users to specify the password length and select the types of characters to include, such as uppercase letters, lowercase letters, numbers, and special symbols. The generated password should meet standard security guidelines and use randomness to enhance password strength.

Algorithm:

Step 1: Start the program.

Step 2: Ask the user to enter the required password length.

Step 3: Ask the user whether to include:

Uppercase letters

Lowercase letters

Numbers

Special symbols

Step 4: Create an empty character set.

Step 5: Add the selected character types to the character set.

Step 6: Check whether the character set is empty.

If Yes, display "Please select at least one character type" and stop.

If No, continue.

Step 7: Randomly select characters from the character set until the required password length is reached.

Step 8: Display the generated password.

Step 9: End the program.

Flowchart:



